

AP Calculus II
Notes 9.7
Taylor and Maclaurin Polynomials

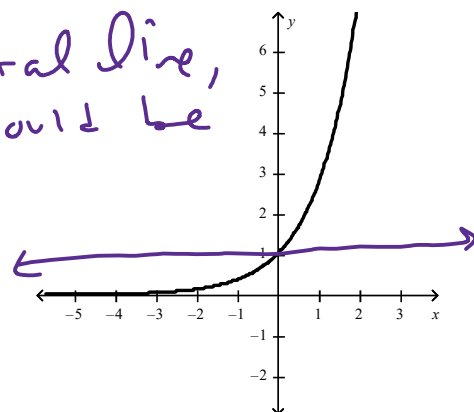
The purpose of this next unit is to show how transcendental and trigonometric functions can be approximated by polynomials.

← $P_n(x) = ax^n + bx^{n-1} + cx^{n-2} + \dots + C$

Let's start off with trying to approximate the curve $f(x) = e^x$ at $x = 0$. The most basic approximation of this function would have a degree of 0 (a constant) and would be:

A constant would be a horizontal line, so the only thing in common would be a y-value (a point).

$P_0(x) = 1$ since $e^0 = 1$



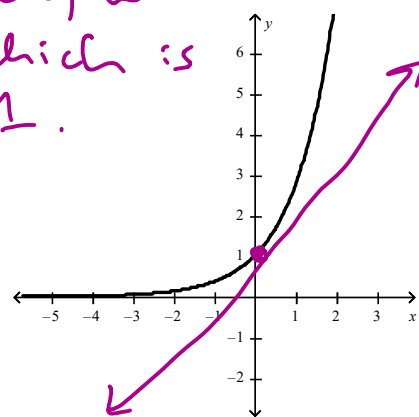
How can we make this approximation better?

Let's add slope into the mix. So, we want point and slope to be the same, which is a tangent line! Degree is now 1.

Point : $(0, 1)$ slope : $f'(x) = e^x$
 $f'(0) = 1$

$y - 1 = 1(x - 0)$

or $P_1(x) = 1 + x \approx e^x$



Check:

	$P_1(x)$	$f(x) = e^x$	
Point	$P_1(0) = 1$	$f(0) = e^0 = 1$	✓
Slope	$P_1'(x) = 1$ $P_1'(0) = 1$	$f'(x) = e^x$ $f'(0) = 1$	✓✓

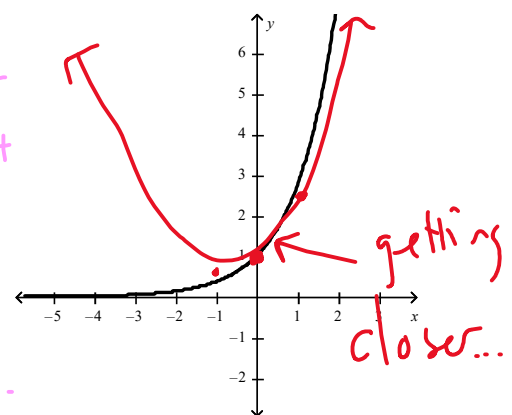
So now we have 2 things in common...

let's add curve to the mix... Concavity!

How can we make *this* approximation any better?

increase the degree to 2!

We want same point, slope + concavity. But we aren't great at parabola writing:



$$P_2(x) = 1 + x + x^2 \leftarrow \text{guess...}$$

Check:

	$P_2(x) = 1 + x + x^2$	$f(x) = e^x$	
pt	$P_2(0) = 1$	$f(0) = 1$	✓
Slope	$P_2'(x) = 1 + 2x$ $P_2'(0) = 1$	$f'(x) = e^x$ $f'(0) = 1$	✓✓
concavity	$P_2''(x) = 2$	$f''(x) = e^x$ $f''(0) = 1$	XX

So the constant and x term is good, but 2nd degree is twice as much... So let's divide by 2!

$$P_2(x) = 1 + x + \frac{x^2}{2} \quad (\text{still guess})$$

	$P_2(x) = 1 + x + \frac{x^2}{2}$	$f(x) = e^x$	
pt	$P_2(0) = 1$	$f(0) = 1$	✓
Slope	$P_2'(x) = 1 + x$, $P_2'(0) = 1$	$f'(x) = e^x$, $f'(0) = 1$	✓✓
concavity	$P_2''(x) = 1$, $P_2''(0) = 1$	$f''(x) = e^x$, $f''(0) = 1$	✓✓✓

yes!! pattern???

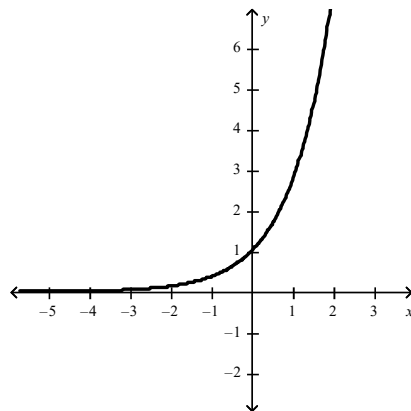
Let's get to a 3rd degree and third derivative!

How can we make this approximation any better?

Write a 3rd degree polynomial that has the same point, slope, concavity, and ... change in concavity

Guess: $P_3(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{3}$

maybe??



$$P_3' = 1 + x + x^2 \quad f' = e^x$$

$$P_3'' = 1 + 2x \quad f'' = e^x$$

$$P_3''' = 2 \quad f''' = e^x$$

Check:

would be same ... so let's divide the $\frac{x^3}{3}$ by 2

$$P_3(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} \leftarrow \text{Guess}$$

$$P_3 = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} \Big|_{x=0} = 1 \quad \checkmark \quad e^0 = 1$$

$$P_3' = 1 + x + \frac{x^2}{2} \Big|_{x=0} = 1 \quad \checkmark \checkmark \quad e^0 = 1$$

$$P_3'' = 1 + x \Big|_{x=0} = 1 \quad \checkmark \checkmark \checkmark \quad e^0 = 1$$

$$P_3''' = 1 \quad = 1 \quad \checkmark \checkmark \checkmark \checkmark \quad e^0 = 1$$

So $P_3(x) = 1 + \frac{x}{1} + \frac{x^2}{2} + \frac{x^3}{6} \approx e^x$
pattern???

Definition of Maclaurin Polynomials:

If $f(x)$ has n derivatives at $c = 0$, then:

$$P_n(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \dots + \frac{f^{(n)}(0)}{n!}x^n$$

ensures same point

continuity

is also known as the **n th Maclaurin polynomial** for $f(x)$ that is centered at $c = 0$.

change of continuity

n^{th} derivative of $f(x)$ at $x=0$

called the general term

Ex. 1: Find the Maclaurin polynomials P_4, P_5 and P_6 for $f(x) = e^x$ and use each to approximate $e^{0.1}$.

$$P_4(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} \approx e^x \quad P_4(0.1) = 1.10517083$$

$$P_5(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} \approx e^x \quad P_5(0.1) = 1.10517091$$

$$P_6(x) = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{120} + \frac{x^6}{720} \approx e^x \quad P_6(0.1) = 1.1051709183$$

$$e^{0.1} = 1.105170918075 \dots \quad \text{accurate!}$$

Ex. 2: Find the Maclaurin polynomial of degree 6 for $f(x) = \cos 2x$. How many nonzero terms are in the polynomial?

$$P_6(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \frac{f^{(6)}(0)}{6!}x^6$$

we need $f(0) \rightarrow f^{(6)}(0)$ for $f(x) = \cos 2x$

$$\begin{array}{lcl} f = \cos 2x & \rightarrow & 1 \\ f' = -2\sin 2x & \rightarrow & 0 \\ f'' = -4\cos 2x & \rightarrow & -4 \\ f''' = 8\sin 2x & \rightarrow & 0 \\ f^{(4)} = 16\cos 2x & \rightarrow & 16 \\ f^{(5)} = -32\sin 2x & \rightarrow & 0 \\ f^{(6)} = -64\cos 2x & \rightarrow & -64 \end{array} \quad x=0$$

$$P_6(x) = 1 + \frac{-4}{2!}x^2 + \frac{16}{4!}x^4 + \frac{-64}{6!}x^6$$

$$P_6(x) = 1 - \frac{4}{2!}x^2 + \frac{16}{4!}x^4 - \frac{64}{6!}x^6 \approx \cos 2x$$

note symbols

there are 4 nonzero terms

not necessarily 4th degree

Ex. 3: Find the first 4 nonzero terms of the Maclaurin polynomial for $g(x) = \frac{6}{2+x}$.

we can assume it goes to $P_3(x)$ and can add more terms if needed:

$$P_3(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3$$

$$\begin{array}{l|l} f = \frac{6}{2+x} & \rightarrow 3:3 \\ f' = \frac{-6}{(2+x)^2} & \rightarrow -6/4 = -3/2 \\ f'' = \frac{12}{(2+x)^3} & \rightarrow 12/8 = 3/2 \\ f''' = \frac{-36}{(2+x)^4} & \rightarrow \frac{-36}{16} = -9/4 \end{array} \quad x=0$$

$$P_3(x) = 3 - \frac{3/2}{1!}x + \frac{3/2}{2!}x^2 - \frac{9/4}{3!}x^3$$

or

$$P_3(x) = 3 - \frac{3}{2}x + \frac{3}{4}x^2 - \frac{3}{8}x^3$$

Ex. 4: Find the Maclaurin polynomial of degree 6 for $f(x) = \cos x + \sin x$. How many nonzero terms are in the polynomial?

must go to 6th derivative, $6!$, x^6

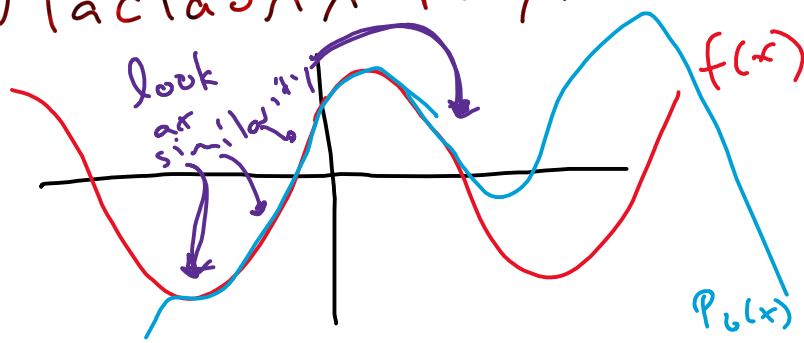
$$P_6(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5 + \frac{f^{(6)}(0)}{6!}x^6$$

$$\begin{array}{l|l} f = \cos x + \sin x & = 1 \\ f' = -\sin x + \cos x & = 1 \\ f'' = -\cos x - \sin x & = -1 \\ f''' = \sin x - \cos x & = -1 \\ f^{(4)} = \cos x + \sin x & = 1 \\ f^{(5)} = -\sin x + \cos x & = 1 \\ f^{(6)} = -\cos x - \sin x & = -1 \end{array} \quad x=0$$

$$P_6(x) = 1 + \frac{1}{1!}x - \frac{1}{2!}x^2 - \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \frac{1}{5!}x^5 - \frac{1}{6!}x^6$$

$$\approx \cos x + \sin x$$

7 terms in 6th degree
Maclaurin Polynomial



What polynomial starts to conform about... "centered" at $x=c$

Definition of Taylor Polynomials:

If at c , then:

$$P_n(x) = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \frac{f'''(c)}{3!}(x-c)^3 + \dots + \frac{f^{(n)}(c)}{n!}(x-c)^n$$

is also known as the **nth Taylor polynomial for $f(x)$** . A Maclaurin Polynomial is a specific type of Taylor Polynomial with $c = 0$.

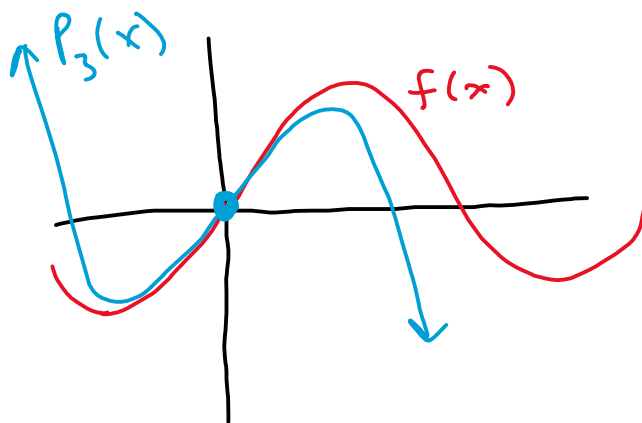
Ex. 5: Find the third-degree Taylor polynomial for $f(x) = 2 \sin x$, centered at the following values:

a) $c = 0$

$$P_3(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3$$

$$\begin{array}{l|l} f = 2\sin x & = 0 \\ f' = 2\cos x & = 2 \\ f'' = -2\sin x & = 0 \\ f''' = -2\cos x & = -2 \end{array}$$

$$P_3(x) = 2x - \frac{2}{3!}x^3$$

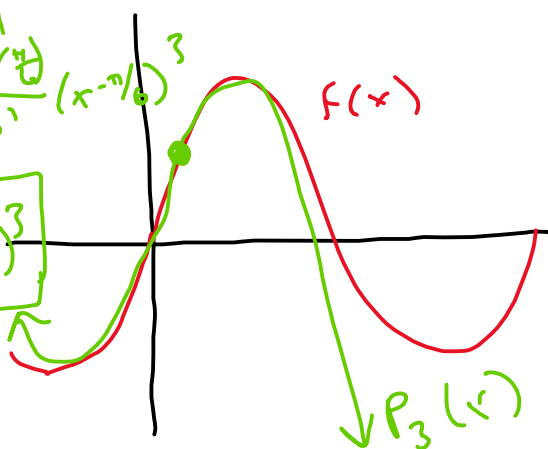


b) $c = \frac{\pi}{6}$

$$P_3(x) = f(\pi/6) + \frac{f'(\pi/6)}{1!}(x-\pi/6) + \frac{f''(\pi/6)}{2!}(x-\pi/6)^2 + \frac{f'''(\pi/6)}{3!}(x-\pi/6)^3$$

$$\begin{array}{l|l} f & = 1 \\ f' & = \sqrt{3} \\ f'' & = -1 \\ f''' & = -\sqrt{3} \end{array}$$

$$P_3(x) = 1 + \sqrt{3}(x-\pi/6) - \frac{1}{2!}(x-\pi/6)^2 - \frac{\sqrt{3}}{3!}(x-\pi/6)^3$$



c) $c = \frac{3\pi}{4}$

$$P_3(x) = f(3\pi/4) + \frac{f'(3\pi/4)}{1!}(x-3\pi/4) + \frac{f''(3\pi/4)}{2!}(x-3\pi/4)^2 + \frac{f'''(3\pi/4)}{3!}(x-3\pi/4)^3$$

$$f = \sqrt{2}$$

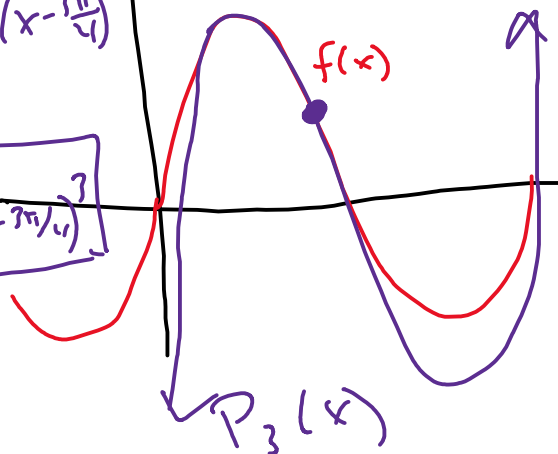
$$f' = -\sqrt{2}$$

$$f'' = -\sqrt{2}$$

$$f''' = \sqrt{2}$$

$$c = 3\pi/4$$

$$P_3(x) = \sqrt{2} - \sqrt{2}(x-3\pi/4) - \frac{\sqrt{2}}{2!}(x-3\pi/4)^2 + \frac{\sqrt{2}}{3!}(x-3\pi/4)^3$$



Ex. 4: Find the Taylor polynomial P_3 for $f(x) = \ln|\sin x|$ at $c = \frac{\pi}{2}$. Then, use P_3 to approximate $f(1.5)$.

$$P_3 = f(\pi/2) + \frac{f'(\pi/2)}{1!}(x-\pi/2) + \frac{f''(\pi/2)}{2!}(x-\pi/2)^2 + \frac{f'''(\pi/2)}{3!}(x-\pi/2)^3$$

$$\begin{array}{l|l} f = \ln|\sin x| & = 0 \\ f' = \frac{1}{\sin x} \cdot \cos x = \cot x & = 0 \\ f'' = -\csc^2 x = \frac{-1}{\sin^2 x} & = -1 \\ f''' = -2\csc x (\csc x \cot x) & = 0 \end{array} \quad \left| \begin{array}{l} \\ \\ \\ c = \pi/2 \end{array} \right.$$

$$P_3(x) = \frac{-1}{2!}(x-\pi/2)^2$$

$$P_3(1.5) = -0.0025060599$$

$$\underline{P_3(1.5) \approx f(1.5)}!! \quad f(1.5) = -0.0025081562$$

Ex. 5: Find the fourth degree term of the Taylor polynomial for $f(x) = x^2 + \ln x$ centered at $c = 1$.

↑
just term!! NOT whole polynomial!

$$\frac{f^{(4)}(1)}{4!}(x-1)^4 \text{ is 4th degree term}$$

$$\begin{array}{l} f = x^2 + \ln x \\ f' = 2x + \frac{1}{x} \\ f'' = 2 - \frac{1}{x^2} \\ f''' = \frac{2}{x^3} \\ f^{(4)} = \frac{-6}{x^4} \Big|_{x=1} = -6 \end{array}$$

$$\frac{-6}{4!}(x-1)^4$$

next derivative = -2 * previous derivative

Ex. 6: Let f be a function such that $f(2) = 4$, $f'(2) = -3$ and $f^{(n+1)}(2) = -2f^{(n)}(2)$, for all $n \geq 1$.

a) Write the first four nonzero terms of the Taylor Polynomial for f about $x = 2$.

$$P_3(x) = f(2) + \frac{f'(2)}{1!}(x-2) + \frac{f''(2)}{2!}(x-2)^2 + \frac{f'''(2)}{3!}(x-2)^3$$

$$f(2) = 4$$

$$f'(2) = -3$$

$$f''(2) = -2f'(2) = 6$$

$$f'''(2) = -12$$

$$P_3(x) = 4 - 3(x-2) + \frac{6}{2!}(x-2)^2 - \frac{12}{3!}(x-2)^3$$

b) Write the fourth degree Taylor Polynomial for g , where $g(x) = f'(x)$ about $x = 2$.

$$P_4(x) = g(2) + \frac{g'(2)}{1!}(x-2) + \frac{g''(2)}{2!}(x-2)^2 + \frac{g'''(2)}{3!}(x-2)^3 + \frac{g^{(4)}(2)}{4!}(x-2)^4$$

$$g(2) = f'(2) = -3, g'(2) = f''(2) = 6, g''(2) = f'''(2) = -12, g'''(2) = f^{(4)}(2) = 24, g^{(4)}(2) = -48$$

$$P_4(x) = -3 + 6(x-2) - \frac{12}{2!}(x-2)^2 + \frac{24}{3!}(x-2)^3 - \frac{48}{4!}(x-2)^4$$

c) Write the third degree Taylor Polynomial for h , where $h(x) = x^3 + \int_2^{3x-4} f(t)dt$ about $x = 2$.

$$P_3(x) = h(2) + \frac{h'(2)}{1!}(x-2) + \frac{h''(2)}{2!}(x-2)^2 + \frac{h'''(2)}{3!}(x-2)^3$$

$$h(2) = 8 + \int_2^2 f(t)dt = 8$$

$$h'(x) = 3x^2 + f(3x-4) \cdot 3, h'(2) = 12 + 3f(2) = 12 + 12 = 24$$

$$h''(x) = 6x + 9f'(3x-4), h''(2) = 12 + 9f'(2) = 12 - 27 = -15$$

$$h'''(x) = 6 + 27f''(3x-4), h'''(2) = 6 + 27f''(2) = 6 + 162 = 168$$

$$P_3(x) = 8 + 24(x-2) - \frac{15}{2!}(x-2)^2 + \frac{168}{3!}(x-2)^3$$

Ex. 7: The third-degree Taylor polynomial for the function f about $x = 0$ is

$T(x) = 3 - 4x + 2x^2 - 3x^3$. Which of the following tables gives the values of f and its first three derivatives at $x = 0$?

(a)

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	3	-8	6	-12

(b)

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	3	-4	2	-3

(c)

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	3	-4	4	-18

(d)

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	3	-4	4	-9

$$T(x) = 3 - 4x + 2x^2 - 3x^3$$

$$3 = f(0) \rightarrow f(0) = 3$$

$$-4x = \frac{f'(0)}{1!}x \rightarrow f'(0) = -4$$

$$2x^2 = \frac{f''(0)}{2!}x^2 \rightarrow f''(0) = 4$$

$$-3x^3 = \frac{f'''(0)}{3!}x^3 \rightarrow f'''(0) = -18$$

Ex. 8: The functions f , f' , and f'' are each continuous and differentiable. The n th derivative of f is

given by $f^{(n)}(1) = \sum_{i=0}^{\infty} 12 \left(\frac{n+1}{5} \right)^i$ when $0 \leq n \leq 3$. Find the third degree Taylor polynomial for $f(x)$ centered around $x = 1$.

$$P_3(x) = f(1) + \frac{f'(1)}{1!}(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \frac{f'''(1)}{3!}(x-1)^3$$

$$f(1) = f^{(0)}(1) = \sum_{i=0}^{\infty} 12 \left(\frac{1}{5} \right)^i \xrightarrow{\text{Geo. Series!}} \frac{12}{1 - 1/5} = \frac{60}{5-1} = 15$$

$$f'(1) = \sum_{i=0}^{\infty} 12 \left(\frac{2}{5} \right)^i = \frac{12}{1 - 2/5} = \frac{60}{5-2} = 20$$

$$f''(1) = \sum_{i=0}^{\infty} 12 \left(\frac{3}{5} \right)^i = \frac{12}{1 - 3/5} = \frac{60}{5-3} = 30$$

$$f'''(1) = \sum_{i=0}^{\infty} 12 \left(\frac{4}{5} \right)^i = \frac{12}{1 - 4/5} = \frac{60}{5-4} = 60$$

$$P_3(x) = 15 + 20(x-1) + \frac{30}{2!}(x-1)^2 + \frac{60}{3!}(x-1)^3$$

AP Calculus II
Notes 9.10
Taylor and Maclaurin Power Series

In the last section, we found that Taylor and Maclaurin polynomials can approximate various functions. For example, we know the Maclaurin polynomials $P_1 \rightarrow P_5$ for the function $f(x) = e^x$ are as follows:

$$P_1 = 1 + x \approx e^x$$

$$P_2 = 1 + x + \frac{x^2}{2!} \approx e^x$$

$$P_3 = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} \approx e^x$$

$$P_4 = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} \approx e^x$$

$$P_5 = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} \approx e^x$$

We also found that the higher the degree of the polynomial, the better the approximation becomes about c . Therefore, if we wanted to find the exact representation of these functions, we use power series.

let's have ∞ many terms added together!

Definition of Power Series

If x is a variable, then an infinite series of the form:

$$\sum_{n=0}^{\infty} a_n(x-c)^n = a_0 + a_1(x-c) + a_2(x-c)^2 + a_3(x-c)^3 + \cdots + a_n(x-c)^n$$

is called a **power series centered at c** and $a_n(x-c)^n$ is called the **general term**.

Taylor and Maclaurin Power Series

If a function $f(x)$ has derivatives of all orders at $x = c$, then the series:

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!} (x-c)^n = f(c) + f'(c)(x-c) + \frac{f''(c)}{2!} (x-c)^2 + \frac{f'''(c)}{3!} (x-c)^3 + \cdots + \frac{f^{(n)}(c)}{n!} (x-c)^n$$

is called the **Taylor series for $f(x)$ centered at c** . Again, a Taylor series centered at $c = 0$ is called a **Maclaurin series**.

this is the general term!

If you can find the pattern for the coefficients of the Taylor polynomials for a function, then you can extend the pattern easily for the corresponding Taylor series.

Ex. 1: Write the fourth-degree Maclaurin polynomial for $f(x) = e^x$, write the general term and the Maclaurin series for $f(x) = e^x$.

center is 0

$$P_4(x) = \frac{x^0}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} \approx e^x$$

the pattern seems to be $\frac{1x^n}{n!}$ Same power & factorial

General term $\rightarrow \frac{x^n}{n!}$

Maclaurin Series: $\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots = e^x$ wow!
 ∞ many terms

Ex. 2: Write the fourth-degree Taylor polynomial for $f(x) = \ln x$, centered about $c = 1$, write the general term and the Taylor series for $f(x) = \ln x$, centered about $c = 1$.

$$P_4(x) = f(1) + \frac{f'(1)}{1!}(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \frac{f'''(1)}{3!}(x-1)^3 + \frac{f^{(4)}(1)}{4!}(x-1)^4$$

$f = \ln x$	$= 0$	$P_4(x) = \frac{1}{1!}(x-1)^1 - \frac{1}{2!}(x-1)^2 + \frac{2}{3!}(x-1)^3 - \frac{6}{4!}(x-1)^4$ $= \frac{1}{1}(x-1)^1 - \frac{1}{2}(x-1)^2 + \frac{1}{3}(x-1)^3 - \frac{1}{4}(x-1)^4$
$f' = \frac{1}{x}$	$= 1$	
$f'' = -\frac{1}{x^2}$	$= -1$	
$f''' = \frac{2}{x^3}$	$= 2$	
$f^{(4)} = -\frac{6}{x^4}$	$= -6$	

pattern: alternates, no factorial

$\frac{(-1)^n (x-1)^n}{n} \rightarrow \ln x = \sum_{n=1}^{\infty} \frac{(-1)^n}{n} (x-1)^n$

(start at $n=1$)

Ex. 3: Write the Maclaurin series for $f(x) = \sin x$.

Write a Maclaurin polynomial to discover the pattern

$$P_5(x) = f(0) + \frac{f'(0)}{1!}x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \frac{f^{(5)}(0)}{5!}x^5$$

$f = \sin x$	0	$P_5(x) = \frac{1}{1!}x^1 - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 + \left(-\frac{1}{7!}x^7 + \frac{1}{9!}x^9 \right)$ $\uparrow \quad \quad \uparrow \quad \quad \uparrow \quad \quad \uparrow$ $n=0 \quad \quad n=1 \quad \quad n=2 \quad \quad n=3 \quad \quad n=4$ if we kept going
$f' = \cos x$	1	
$f'' = -\sin x$	0	
$f''' = -\cos x$	-1	
$f^{(4)} = \sin x$	0	
$f^{(5)} = \cos x$	1	
$f^{(6)} = -\sin x$	0	

$f^{(7)} = -\cos x = -1$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

Ex. 4: Write the Maclaurin series for $f(x) = \cos x$.

$f = \cos x$	1	$P_8(x) = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \frac{1}{8!}x^8$ $n=0 \quad n=1 \quad n=2 \quad n=3 \quad n=4$ $\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$ * note. $2n! \neq (2n)!$
$f' = -\sin x$	0	
$f'' = -\cos x$	-1	
$f''' = \sin x$	0	
$f^{(4)} = \cos x$	1	
$f^{(5)} = -\sin x$	0	
$f^{(6)} = -\cos x$	-1	
$f^{(7)} = \sin x$	0	
$f^{(8)} = \cos x$	1	
$x=0$		

The Maclaurin Series You MUST Know:

expans form

sigma form

$$e^0 = 1 + 0 + 0 = 1 \quad 1) \quad e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots + \frac{x^n}{n!} = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$\sin 0 = 0 - 0 + 0 = 0 \quad 2) \quad \sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots - \frac{(-1)^n x^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\cos 0 = 1 - 0 + 0 = 1 \quad 3) \quad \cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots - \frac{(-1)^n x^{2n}}{(2n)!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

$$\frac{1}{1-0} = 1 + 0 + 0 = 1 \quad 4) \quad \frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \dots + x^n = \sum_{n=0}^{\infty} x^n$$

↑ geometric!

Ex. 5: Find the Maclaurin power series for $f(x) = \sin(x^2)$.

$f = \sin x^2$

$f' = 2x \cos x^2$

$f'' = 2 \cos x^2 - 4x^2 \sin x^2$

crazy more derivatives needed

Instead:

$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots - \frac{(-1)^n x^{2n+1}}{(2n+1)!}$

$\sin x^2 = x^2 - \frac{(x^2)^3}{3!} + \frac{(x^2)^5}{5!} - \dots - \frac{(-1)^n (x^2)^{2n+1}}{(2n+1)!}$

$\sin x^2 = x^2 - \frac{x^6}{3!} + \frac{x^{10}}{5!} - \dots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+2}}{(2n+1)!}$

Ex. 6: Find the Taylor power series for $f(x) = \frac{e^{(-3x)} - 1}{3}$ centered about $x = 0$.

Let's Build $f(x)$ from e^x

$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$

replace x 's with $-3x$

$e^{-3x} = 1 - 3x + \frac{9x^2}{2!} - \frac{27x^3}{3!} + \dots + \frac{(-3x)^n}{n!}$

$e^{-3x} - 1 = -3x + \frac{9x^2}{2!} - \frac{27x^3}{3!} + \dots + \frac{(-3x)^n}{n!}$

Subtract 1 from both sides

starting at $n=1$

Since we removed $n=0$

$\frac{e^{-3x} - 1}{3} = -x + \frac{3x^2}{2!} - \frac{9x^3}{3!} + \dots + \frac{(-3x)^n}{n! \cdot 3} = f(x) = \sum_{n=1}^{\infty} \frac{(-3x)^n}{3n!}$

let's write out 4 in middle steps to be sure

Ex. 7: Find the first three nonzero terms in the Maclaurin series $g(x) = x \cos x - \sin x$.

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \quad \text{no need for general term}$$

$$x \cos x = x - \frac{x^3}{2!} + \frac{x^5}{4!} - \frac{x^7}{6!} + \dots$$

$$- \left(\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \right) \quad \text{only add + subtract like terms + need common denominator!}$$

$$x \cos x - \sin x = g(x) = \left(\frac{-1}{2!} + \frac{1}{3!} \right) x^3 + \left(\frac{1}{4!} - \frac{1}{5!} \right) x^5 + \left(\frac{-1}{6!} + \frac{1}{7!} \right) x^7 + \dots$$

$$= -\frac{2}{3!} x^3 + \frac{4}{5!} x^5 - \frac{6}{7!} x^7 + \dots$$

Ex. 8: Evaluate the following infinite series:

a) $\sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n}}{3^{2n} (2n)!} = \sum_{n=0}^{\infty} \frac{(-1)^n (\pi/3)^{2n}}{(2n)!}$

b) $4 - \frac{4x^2}{3!} + \frac{4x^4}{5!} - \frac{4x^6}{7!} + \dots$

focus on factorial looks like cosine

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} \quad \leftarrow \text{replaces } x \text{ with } \pi/3$$

So, $\sum_{n=0}^{\infty} \frac{(-1)^n \pi^{2n}}{3^{2n} (2n)!} = \cos \pi/3 = \boxed{\frac{1}{2}}$

c) $\frac{1}{1!} - \frac{2}{2!} + \frac{4}{3!} - \frac{8}{4!} + \frac{16}{5!} - \frac{32}{6!} + \dots$

factorials look like e^x

sigma notation: $\sum_{n=0}^{\infty} \frac{(-1)^n 2^n}{n!}$

$$\sum_{n=0}^{\infty} \frac{(-2)^n}{n!} : e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

So replace x with -2

$$\boxed{1 - 2 + \frac{4}{2!} - \frac{8}{3!} + \frac{16}{4!} - \frac{32}{5!} + \dots = e^{-2}}$$

Factorials look like $\sin x$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots \quad \bullet 4$$

$$4 \sin x = 4x - \frac{4x^3}{3!} + \frac{4x^5}{5!} - \frac{4x^7}{7!} + \dots \quad \div x$$

$$\boxed{\frac{4 \sin x}{x} = 4 - \frac{4x^2}{3!} + \frac{4x^4}{5!} - \frac{4x^6}{7!} + \dots}$$

d) $1 - \frac{x^4}{2!} + \frac{x^8}{4!} - \frac{x^{12}}{6!} + \dots$

factorial looks like $\cos x$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

powers are doubled

$$\cos(x^2) = 1 - \frac{(x^2)^2}{2!} + \frac{(x^2)^4}{4!} - \frac{(x^2)^6}{6!} + \dots$$

$$\boxed{\cos(x^2) = 1 - \frac{x^4}{2!} + \frac{x^8}{4!} - \frac{x^{12}}{6!} + \dots}$$

Ex. 9: Given $f(x) = 2x - \frac{2x^3}{3!} + \frac{2x^5}{5!} - \frac{2x^7}{7!} + \dots + \frac{(-1)^n 2x^{2n+1}}{(2n+1)!}$, $\leftarrow 2$ isn't raised to any power, so the 2 isn't with the x

a) Find one value of x for which $f(x) = 1$.

can't solve $1 = 2x - \frac{2x^3}{3!} + \frac{2x^5}{5!} - \frac{2x^7}{7!} + \dots$ as polynomial
So, let's rewrite polynomial as function
 $f(x) = 2 \sin x$ $2 \sin x = 1$, $\sin x = 1/2$ at $x = \pi/6$

b) Find the first 4 non-zero terms for $f'(x)$. Write the Maclaurin series for $f'(x)$?

$f'(x)$ is multiple constants and power rules
 $f'(x) = 2 - \frac{6x^2}{3!} + \frac{10x^4}{5!} - \frac{14x^6}{7!} + \dots + \frac{(-1)^n 2(2n+1)x^{2n}}{(2n+1)!}$ $\leftarrow$ multiply power & subtract 1
 $f'(x) = 2 - \frac{2x^2}{2!} + \frac{2x^4}{4!} - \frac{2x^6}{6!} + \dots + \frac{(-1)^n 2x^{2n}}{(2n)!}$ $\leftarrow$ divided a factor in numerator & had 1 less factorial
 $f'(x) = 2 \cos x$ $\leftarrow$ if $f = 2 \sin x$, $f' = 2 \cos x$

c) If $g'(x) = f(x)$ and $g(0) = -2$, find the first 4 non-zero terms and the general term of $g(x)$.

$g' = 2x - \frac{2x^3}{3!} + \frac{2x^5}{5!} - \frac{2x^7}{7!} + \dots + \frac{(-1)^n 2x^{2n+1}}{(2n+1)!}$ $\leftarrow$ integrate
 $g = x^2 - \frac{2x^4}{4 \cdot 3!} + \frac{2x^6}{6 \cdot 5!} - \frac{2x^8}{8 \cdot 7!} + \dots + \frac{(-1)^n 2x^{(2n+2)}}{(2n+2)(2n+1)!} + C$
 $g = C + x^2 - \frac{2x^4}{4!} + \frac{2x^6}{6!} - \frac{2x^8}{8!} + \dots + \frac{(-1)^n 2x^{(2n+2)}}{(2n+2)!}$

d) Find $f^{(19)}(0)$.

19th degree term is

$-2 = C + 0 - 0 + 0 - 0 + \dots$ $C = -2$
 $g = -2 + x^2 - \frac{2x^4}{4!} + \frac{2x^6}{6!} - \frac{(-1)^n 2x^{2n}}{(2n)!}$

$\frac{f^{(19)}(0)}{19!} x^{19} \rightarrow$ general term is $\frac{(-1)^n 2x^{(2n+1)}}{(2n+1)!}$
uses $n=9$

$\frac{(-1)^9 2x^{19}}{19!} \rightarrow$ so $f^{(19)}(0) = (-1)^9 \cdot 2 = -2$

AP Calculus II

Notes 9.8

Power Series

In the last section, we dealt with Maclaurin and Taylor polynomials that when added up, resulted in Maclaurin and Taylor series. In this section, we will focus on any power series and its features.

Definition of Power Series

If x is a variable, then an infinite series of the form:

$$\sum_{n=0}^{\infty} a_n(x - c)^n = a_0 + a_1(x - c) + a_2(x - c)^2 + a_3(x - c)^3 + \cdots + a_n(x - c)^n$$

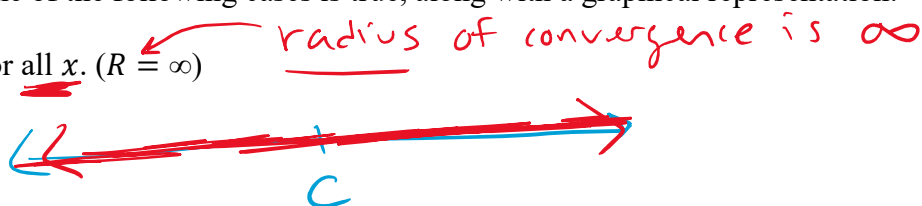
is called a **power series centered at c** .

When representing a function $f(x)$ as a power series, the **domain** of the function $f(x)$ is all the values of x for which the power series converges. The Geometric, Alternating Series and Ratio Tests are very helpful in these situations, but the other tests are just as useful. The set of all x -values for which the power series converges is known as the **interval of convergence**.

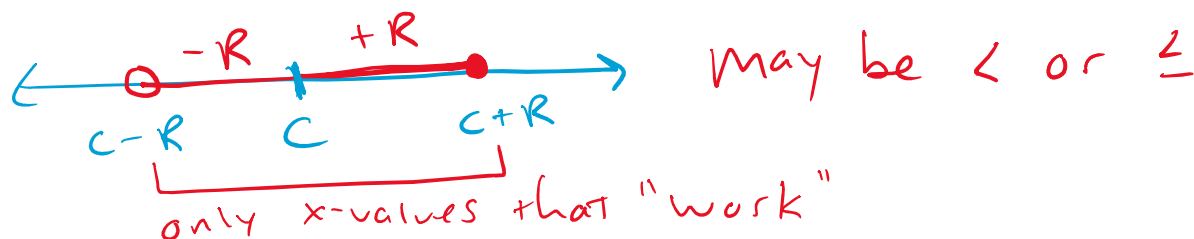
Interval of Convergence of a Power Series:

For a power series centered at c , one of the following cases is true, along with a graphical representation:

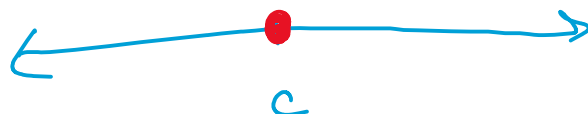
- a) The series converges for all x . ($R = \infty$)



- b) The series converges for only a finite window $|x - c| < R$, with c being the center of this window and R being the **radius of convergence**. Think of R as being the “plus or minus” away from the center. This interval of convergence can include one, none or both endpoints.



- c) The series converges at one point, namely c . ($R = 0$)



only x -value that works is at $x = c$

Use Ratio or Geometric Test ... which converges if $|r| < 1$

Ex. 1: Find the center, radius and interval of convergence for the following power series:

a) $\sum_{n=1}^{\infty} \frac{x^n}{n}$ $\lim_{n \rightarrow \infty} \left| \frac{x^{n+1}}{n+1} \cdot \frac{n}{x^n} \right| < 1 \rightarrow \lim_{n \rightarrow \infty} \left| x \cdot \frac{n}{n+1} \right| < 1$

as $n \rightarrow \infty$, this goes to 1

$|x| < 1 \rightarrow -1 < x < 1$

center is $\boxed{0}$
since $x=0$

so radius is $\boxed{1}$

Since there is a constant radius, we must test the endpoints

$x = -1$: $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ which converges because it's alternating harmonic

$x = 1$: $\sum_{n=1}^{\infty} \frac{(1)^n}{n} = \sum_{n=1}^{\infty} \frac{1}{n}$ which is divergent harmonic

So, $\boxed{-1 \leq x < 1}$

b) $\sum_{n=1}^{\infty} n! x^n$ $\lim_{n \rightarrow \infty} \left| \frac{(n+1)! x^{n+1}}{n! x^n} \right| < 1$, $\lim_{n \rightarrow \infty} \left| (n+1) \cdot x \right| < 1$

as $n \rightarrow \infty$, $n+1 \rightarrow \infty$, which is never less than 1.

Center: $C = 0$

radius: $R = 0$

So the only value that would cause the series to converge is $x = \text{center}$

$\boxed{x = 0}$

c) $\sum_{n=3}^{\infty} (-1)^{n+1} \frac{(x+3)^n}{(n-2)^3}$

$\lim_{n \rightarrow \infty} \left| \frac{(x+3)^{n+1}}{(n-1)^3} \cdot \frac{(n-2)^3}{(x+3)^n} \right| < 1 \rightarrow \lim_{n \rightarrow \infty} \left| (x+3) \frac{(n-2)^3}{(n-1)^3} \right| < 1$

as $n \rightarrow \infty$, this goes to 1

$|x+3| < 1$

$-1 < x+3 < 1$

$-4 < x < -2$

center: $x = -3$

radius: $R = 1$

Test Endpoints: see if $x = -4$ or $x = -2$ converge

$x = -4$: $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} \cdot (-1)^n}{(n-2)^3} = \sum_{n=1}^{\infty} \frac{(-1)^{2n+1}}{(n-2)^3}$ doesn't alternate
 $\sum_{n=1}^{\infty} \frac{1}{(n-2)^3}$ converges to compare to $p=3$

$x = -2$: $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{(n-2)^3}$ converges by alternating series $[-4, -2]$

Ex. 2: Write out the first 4 nonzero terms and the general term of the Maclaurin series for $f(x) = \sin x$. Then, determine the interval of convergence for this function.

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots + \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\lim_{n \rightarrow \infty} \left| \frac{x^{2n+3}}{(2n+3)!} \cdot \frac{(2n+1)!}{x^{2n+1}} \right| < 1 \rightarrow \lim_{n \rightarrow \infty} \left| x^2 \cdot \frac{1}{(2n+3)(2n+2)} \right| < 1$$

as $n \rightarrow \infty$, this goes to 0

$|x^2 \cdot 0| < 1 \rightarrow 0 < 1$ which is always true, so

Interval of convergence: $(-\infty, \infty)$

Ex. 3: The Taylor series for the function $f(x) = \ln x$ is $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} (x-1)^n}{n}$. Find the interval of

convergence for this function. Then, find the value of $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$.

I_{OC}: $\lim_{n \rightarrow \infty} \left| \frac{(x-1)^{n+1}}{n+1} \cdot \frac{n}{(x-1)^n} \right| < 1 \rightarrow \lim_{n \rightarrow \infty} \left| (x-1) \cdot \frac{n}{n+1} \right| < 1$

as $n \rightarrow \infty$, this goes to 1

$|x-1| < 1$ $x=0$: $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} (-1)^n}{n} \leftarrow$ not alternating harmonic $\rightarrow$ Diverges

$-1 < x-1 < 1$ $x=2$: $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} 1^n}{n} \leftarrow$ alternating harmonic $\rightarrow$ Converges

$0 < x < 2$

I_{OC}: $(0, 2]$

$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} \rightarrow \sum_{n=1}^{\infty} \frac{(-1)^{n-1} (x-1)^n}{n}$ means $(x-1)^n = 1$ which is when $x=2$

So, since $x=2$ is in I_{OC} $\rightarrow \ln x = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} (x-1)^n}{n}$

$\ln 2 = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ the alternating harmonic series adds up to $\ln 2$

Ex. 4: The series $\sum_{n=1}^{\infty} a_n (x-2)^n$ converges conditionally when $x=0$. Determine if the following statements are Always true, Sometimes true or Never true.

center is $x=2$

one of the endpoints so $R=2$

a.) The series diverges when $x=5$. Always true Since beyond $R=2$	b.) The series converges when $x=-1$. Never true Since before $x=0$
c.) The series diverges when $x=1$. Never true Inside interval of convergence	d.) The series converges conditionally when $x=4$. Sometimes true the other endpoint so either converges or diverges

Ex. 5: $\sum_{n=1}^{\infty} b_n = -\frac{4}{3} + \frac{4 \cdot 2}{9}(x-2) - \frac{4 \cdot 3}{27}(x-2)^3 + \dots + (-1)^n \frac{4n}{3^n}(x-2)^{n-1} + \dots$

Find the interval of convergence for the series $\sum_{n=1}^{\infty} \frac{b_n}{n^2} = \sum_{n=1}^{\infty} \frac{(-1)^n 4 (x-2)^{n-1}}{3^n n^2}$

$$= \sum_{n=1}^{\infty} \frac{(-1)^n 4 (x-2)^{n-1}}{n 3^n}$$

ROC: $\lim_{n \rightarrow \infty} \left| \frac{4(x-2)^n}{(n+1)3^{n+1}} \cdot \frac{n3^n}{4(x-2)^{n-1}} \right| < 1 \rightarrow \lim_{n \rightarrow \infty} \left| \frac{x-2}{3} \cdot \frac{n}{n+1} \right| < 1$

as $n \rightarrow \infty$, this goes to 1

$\left| \frac{x-2}{3} \right| < 1$

$-1 < \frac{x-2}{3} < 1$

$-3 < x-2 < 3$

$-1 < x < 5$

$x=-1$: $\sum_{n=1}^{\infty} \frac{(-1)^n 4 (-3)^{n-1}}{n 3^n} = \sum_{n=1}^{\infty} \frac{(-1)^n 4 (3)^{n-1} (-1)^{n-1}}{n 3^n}$ not alternating!

$= \sum_{n=1}^{\infty} \frac{4}{3n}$ is **divergent** harmonic

$x=5$: $\sum_{n=1}^{\infty} \frac{(-1)^n 4 (3)^{n-1}}{n 3^n} = \sum_{n=1}^{\infty} \frac{(-1)^n \cdot 4}{3n}$ is alternating harmonic, so **converges**

$-1 < x \leq 5$

Power Series Manipulation

Since these series represent different functions, then we can manipulate the series the same way we manipulate functions. This allows us to differentiate, integrate, find extrema, points of inflection, etc...

Ex. 6: Given $\sum_{n=0}^{\infty} \frac{(-1)^n x^n}{n!}$, find the following:

a) The first four terms and the general term for $f'(x)$. Then, find $f'(0)$ and $f^{(11)}(0)$.

$$f = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots + \frac{(-1)^n x^n}{n!}$$

$$f' = -1 + \frac{2x}{2!} - \frac{3x^2}{3!} + \frac{4x^3}{4!} - \dots + \frac{(-1)^n n x^{n-1}}{n!}$$

$$f' = -1 + x - \frac{x^2}{2!} + \frac{x^3}{3!} - \dots + \frac{(-1)^n x^{n-1}}{(n-1)!}$$

$$f'(0) = -1$$

using $n=12 \dots$

$$+ \frac{x^4}{11!} \rightarrow f^{(11)}(0) = 1$$

b) The first four terms and the general term for $\int f(x) dx$.

$$\int \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots + \frac{(-1)^n x^n}{n!} \right) dx$$

$$= x - \frac{x^2}{2} + \frac{x^3}{3 \cdot 2!} - \frac{x^4}{4 \cdot 3!} + \dots + \frac{(-1)^n x^{n+1}}{(n+1)n!} + C$$

$$= C + x - \frac{x^2}{2!} + \frac{x^3}{3!} - \frac{x^4}{4!} + \dots + \frac{(-1)^n x^{n+1}}{(n+1)!}$$

c) The first four terms and the general term for $g'(t)$ given $g(t) = f(t^3)$.

$$f(x) = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots + \frac{(-1)^n x^n}{n!}$$

replace x with t^3

$$f(t^3) = 1 - t^3 + \frac{t^6}{2!} - \frac{t^9}{3!} + \frac{t^{12}}{4!} - \dots + \frac{(-1)^n t^{3n}}{n!} = g(t)$$

$$g'(t) = 0 - 3t^2 + \frac{6t^5}{2!} - \frac{9t^8}{3!} + \frac{12t^{11}}{4!} - \dots + \frac{(-1)^n 3n(t)^{3n-1}}{n!}$$

$$g' = -3t^2 + \frac{3t^5}{1!} - \frac{3t^8}{2!} + \frac{3t^{11}}{3!} - \dots + \frac{(-1)^n 3t^{3n-1}}{(n-1)!}$$

Ex. 7: Given $f(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(x-1)^{n+1}}{(n+1)(n+2)}$, find the interval of convergence for $f'(x)$.

$f'(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} \cancel{(n+1)} (x-1)^{\cancel{n}} (1)}{\cancel{(n+1)}(n+2)} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} (x-1)^n}{n+2}$

IOC: $\lim_{n \rightarrow \infty} \left| \frac{(x-1)^{n+1}}{n+3} \cdot \frac{n+2}{(x-1)^n} \right| < 1 \rightarrow \lim_{n \rightarrow \infty} \left| (x-1) \cdot \frac{n+2}{n+3} \right| < 1$
 as $n \rightarrow \infty$, this goes to 1

$|x-1| < 1$
 $-1 < x-1 < 1$
 $0 < x < 2$

$x=0$: $\sum \frac{(-1)^{n+1}(-1)^n}{n+2} \leftarrow$ not alternating harmonic, so diverges

$x=2$: $\sum \frac{(-1)^{n+1}(1)^n}{n+2} \leftarrow$ alternating harmonic, so converges

$(0, 2]$

Ex. 8: Find the first 3 nonzero terms and the general term for $\int_0^x \sin(t^4) dt$.

u-sub doesn't work....

$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots + \frac{(-1)^n x^{2n+1}}{(2n+1)!}$

$\sin t^4 = t^4 - \frac{t^{12}}{3!} + \frac{t^{20}}{5!} - \frac{t^{28}}{7!} + \dots + \frac{(-1)^n t^{8n+4}}{(2n+1)!}$

$\int_0^x \sin t^4 dt = \int_0^x \left[t^4 - \frac{1}{3!} t^{12} + \frac{1}{5!} t^{20} - \frac{1}{7!} t^{28} + \dots + \frac{(-1)^n}{(2n+1)!} t^{8n+4} \right] dt$

Can't integrate

definitely can integrate!

$\int_0^x \sin t^4 dt = \left. \frac{1}{5} t^5 - \frac{1}{3! \cdot 13} t^{13} + \frac{1}{5! \cdot 21} t^{21} - \frac{1}{7! \cdot 29} t^{29} + \dots + \frac{(-1)^n}{(2n+1)! \cdot (8n+5)} t^{8n+5} \right|_0^x$

$= \frac{1}{5} x^5 - \frac{1}{13 \cdot 3!} x^{13} + \frac{1}{21 \cdot 5!} x^{21} - \dots + \frac{(-1)^n}{(8n+5)(2n+1)!} x^{8n+5} - (0-0+0)$

1 2 3 terms

Ex. 9: Given $f(x) = \frac{\cos x - 1}{x^2}$, find the following using series:

a) $\lim_{x \rightarrow 0} f(x)$, then verify using L'Hôpital's rule.

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots \quad / \quad \cos x - 1 = -\frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$$

$$\frac{\cos x - 1}{x^2} = -\frac{1}{2!} + \frac{x^2}{4!} - \frac{x^4}{6!} + \dots = f(x)$$

$$\lim_{x \rightarrow 0} \frac{\cos x - 1}{x^2} = -\frac{1}{2!} + \frac{0^2}{4!} - \frac{0^4}{6!} + \dots = -\frac{1}{2} + 0 - 0 + \dots = \boxed{-\frac{1}{2}}$$

L'Hôpital $\rightarrow \lim_{x \rightarrow 0} \frac{\cos x - 1}{x^2} = \frac{0}{0} = \lim_{x \rightarrow 0} \frac{-\sin x}{2x} = \frac{0}{0} = \lim_{x \rightarrow 0} \frac{-\cos x}{2} = -\frac{1}{2}$

b) First 4 nonzero terms and the general term for $\int f(x) dx$.

$$f(x) = -\frac{1}{2!} + \frac{x^2}{4!} - \frac{x^4}{6!} + \frac{x^6}{8!} - \dots + \frac{(-1)^n x^{2n-2}}{(2n)!}$$

$$\int f(x) dx = C + \frac{-1}{2!} x^1 + \frac{x^3}{3 \cdot 4!} - \frac{x^5}{5 \cdot 6!} + \frac{x^7}{7 \cdot 8!} + \dots + \frac{(-1)^n x^{2n+1}}{(2n+1)(2n)!}$$

Ex. 10: Given $g(x) = \frac{e^x - e^{-x}}{2}$, find:

a) The power series expansion of $g(x)$ by showing the first 4 nonzero terms and general term.

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots + \frac{x^n}{n!}$$
$$-(e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \frac{x^5}{5!} + \dots + \frac{(-x)^n}{n!} \leftarrow \frac{(-1)^n x^n}{n!})$$

$$e^x - e^{-x} = 2x + \frac{2x^3}{3!} + \frac{2x^5}{5!} + \frac{2x^7}{7!} + \dots + \frac{2x^{2n+1}}{(2n+1)!}$$

$$g(x) = \frac{e^x - e^{-x}}{2} = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \dots + \frac{x^{2n+1}}{(2n+1)!}$$

b) The relationship between $g(x)$ and the series $h(x) = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots$

$$g' = 1 + \frac{3x^2}{3!} + \frac{5x^4}{5!} + \frac{7x^6}{7!} + \dots$$

$$g' = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots = h(x)$$

c) The equation represented in the power series expansion for the function $h(x)$.

Since $h = g'$ according to the series expansion,
 $h = g'$ in the equation of the function

$$h = g' = \frac{1}{2}(e^x + e^{-x})$$

Ex. 11:

The function f is defined by the power series

$$f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n+1)!} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \dots + \frac{(-1)^n x^{2n}}{(2n+1)!} + \dots$$

for all real numbers x .

- (a) Find $f'(0)$ and $f''(0)$. Determine whether f has a local maximum, a local minimum, or neither at $x = 0$. Give a reason for your answer.
- (c) Show that $y = f(x)$ is a solution to the differential equation $xy' + y = \cos x$.

$$a) f'(x) = -\frac{2x}{3!} + \frac{4x^3}{5!} - \frac{6x^5}{7!} + \dots + \frac{(-1)^n (2n) x^{2n-1}}{(2n+1)!}$$

$$f'(0) = -0 + 0 - 0 + \dots = 0, \quad \boxed{f'(0) = 0}$$

$$f''(x) = -\frac{2}{3!} + \frac{12x^2}{5!} - \frac{30x^4}{7!} + \dots - \frac{(-1)^n (2n)(2n-1) x^{2n-2}}{(2n+1)!}$$

$\boxed{f''(0) = -2/3!}$ Since $f'(0) = 0$ and $f''(0) < 0$, f has a critical value and is concave down. Rel Max

c) Build $xy' + y$ to show the expansion is $\cos x$

$$y' = -\frac{2}{3!}x + \frac{4x^3}{5!} - \frac{6x^5}{7!} + \dots + \frac{(-1)^n (2n) x^{2n-1}}{(2n+1)!}$$

$$xy' = -\frac{2}{3!}x^2 + \frac{4x^4}{5!} - \frac{6x^6}{7!} + \dots + \frac{(-1)^n (2n) x^{2n}}{(2n+1)!}$$

$$+ y = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \dots + \frac{(-1)^n x^{2n}}{(2n+1)!}$$

$$xy' + y = 1 - \frac{3x^2}{3!} + \frac{5x^4}{5!} - \frac{7x^6}{7!} + \dots + \frac{(-1)^n (2n+1) x^{2n}}{(2n+1)!}$$

$$\boxed{xy' + y = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots + \frac{(-1)^n x^{2n}}{(2n)!} = \cos x!}$$

Ex. 12: The function f has derivatives of all orders at $x=0$, and the Maclaurin series for f is

$$\sum_{n=2}^{\infty} \frac{\ln n}{3^n n^3} x^n = \frac{\ln 2}{9 \cdot 8} x^2 + \frac{\ln 3}{27 \cdot 27} x^3 + \frac{\ln 4}{81 \cdot 64} x^4 + \dots$$

- (a) Find $f'(0)$ and $f^{(4)}(0)$.
 (b) Does f have a relative minimum, a relative maximum, or neither at $x=0$? Justify your answer.
 (c) Using the ratio test, determine the interval of convergence of the Maclaurin series for f . Justify your answer.

a) There is no f' term, so $f'(0) = 0$

$f^{(4)}(0)$: 4th degree term is $\frac{f^{(4)}(0)}{4!} x^4 = \frac{\ln 4}{81 \cdot 64} x^4$

So, $f^{(4)}(0) = \frac{4! \cdot \ln 4}{81 \cdot 64}$

b) Since $f'(0) = 0$ and $f''(0) = \frac{2! \cdot \ln 2}{9 \cdot 8} > 0$, then f has a relative min

c) $\lim_{n \rightarrow \infty} \left| \frac{\ln(n+1) x^{n+1}}{3^{n+1} (n+1)^3} \cdot \frac{3^n n^3}{\ln n \cdot x^n} \right| < 1$

$= \lim_{n \rightarrow \infty} \left| \frac{n^3 \ln(n+1)}{(n+1)^3 \ln n} \cdot \frac{x}{3} \right| < 1, \quad \left| \frac{x}{3} \right| < 1, \quad -1 < \frac{x}{3} < 1, \quad -3 < x < 3$

as $n \rightarrow \infty$, this goes to 1

$-3 \leq x \leq 3$

$x = -3$: $\sum \frac{\ln n \cdot (-3)^n}{3^n \cdot n^3} = \sum \frac{(-1)^n \ln n}{n^3}$ i) $\lim_{n \rightarrow \infty} \frac{\ln n}{n^3} = 0$ ✓
 ii) $\frac{\ln(n+1)}{(n+1)^3} < \frac{\ln n}{n^3}$ ✓

$\therefore$ converges by alternating series test

$x = 3$: $\sum \frac{\ln n \cdot 3^n}{3^n \cdot n^3} = \sum \frac{\ln n}{n^3} < \sum \frac{n}{n^3} = \sum \frac{1}{n^2}$ converges by direct comparison to $p=2$ series

AP Calculus II
Notes 9.9
Geometric Power Series

Consider the series $\sum_{n=0}^{\infty} x^n$. We know that this sum can be written as:

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + x^3 + x^4 + \dots + x^n$$

This expression closely resembles the sum of a geometric series with $a = 1$ and $r = x$. Therefore,

the sum of the series would be: $\sum_{n=0}^{\infty} x^n = \frac{a_1}{1-r} = \frac{1}{1-x}$

We can now express functions that look like the sum of a geometric sequence as a **geometric power series** using the general representation that has a center at c :

$$\sum_{n=0}^{\infty} a(x-c)^n = a + a(x-c) + a(x-c)^2 + a(x-c)^3 + \dots = \frac{a}{1-(x-c)}$$

Ex. 1: Write the following expressions as a geometric power series and then write the first 4 terms:

a) $\frac{12}{1-\frac{1}{4}} = \frac{12}{\frac{3}{4}} = 16$

$$\sum_{n=0}^{\infty} 12\left(\frac{1}{4}\right)^n = 16$$

$$= 12 + 3 + \frac{3}{4} + \frac{3}{16} + \dots$$

$$= 16$$

b) $\frac{3}{1-(x+2)} = \frac{3}{-1-x}$

$$\sum_{n=0}^{\infty} 3(x+2)^n$$

$$= 3 + 3(x+2) + 3(x+2)^2 + 3(x+2)^3 + \dots$$

c) $\frac{5x^2}{1+x} = \frac{5x^2}{1-(-x)}$

$$\sum_{n=0}^{\infty} 5x^2(-x)^n$$

$$= \sum_{n=0}^{\infty} (-1)^n 5x^{n+2}$$

$$= 5x^2 - 5x^3 + 5x^4 - 5x^5 + \dots$$

need $\frac{a}{1-r}$ form!

Ex. 2: Find a power series for $f(x) = \frac{4}{x+5}$, centered at the following values. Then, find the interval of convergence for each power series.

a) $c = 0$

$$f(x) = \frac{4}{5+x} = \frac{4/5}{1 + \frac{x}{5}} = \frac{4/5}{1 - (-\frac{x}{5})}$$

center at $c=0$ since just x .

$$\frac{4}{x+5} = \sum_{n=0}^{\infty} \frac{4}{5} \left(-\frac{x}{5}\right)^n = \sum_{n=0}^{\infty} \frac{4}{5} (-1)^n \left(\frac{x}{5}\right)^n = \frac{4}{5} - \frac{4x}{25} + \frac{4x^2}{125} - \dots$$

This series (and being equal to $\frac{4}{x+5}$) converges only during its interval of convergence. Since this is geometric, the series converges when $|r| < 1$. $r = x/5$ here

$$\left|\frac{x}{5}\right| < 1, -1 < \frac{x}{5} < 1 \rightarrow \boxed{-5 < x < 5} \text{ \& no need}$$

b) $c = 3$

to test endpoints since $r=1$ diverges always +

$$f = \frac{4}{5+x} \xrightarrow[\text{centered } -3]{\text{make}} \frac{4}{5+3+(x-3)} = \frac{4}{8+(x-3)} = \frac{4/8}{1 + \frac{x-3}{8}} = \frac{1/2}{1 - \left(-\frac{x-3}{8}\right)}$$

$$\frac{4}{x+5} = \sum_{n=0}^{\infty} \frac{1}{2} \left(\frac{x-3}{-8}\right)^n = \sum_{n=0}^{\infty} \frac{1}{2} (-1)^n \left(\frac{x-3}{8}\right)^n = \frac{1}{2} - \frac{x-3}{16} + \frac{(x-3)^2}{128} - \dots$$

I.o.C.: $\left|\frac{x-3}{8}\right| < 1$ (absolute value takes care of $(-1)^n$)

$$-1 < \frac{x-3}{8} < 1 \rightarrow -8 < x-3 < 8 \rightarrow \boxed{-5 < x < 11}$$

So, $\frac{4}{x+5} = \sum_{n=0}^{\infty} \frac{1}{2} (-1)^n \left(\frac{x-3}{8}\right)^n$ as long as x is between $-5 < x < 11$. Notice -5 is also a domain restriction... coincidence??

Ex. 3: Evaluate $\int \frac{6x}{1-x^3} dx$ using power series centered at 0.

← U-sub won't work...no partial fractions, etc...

But we can integrate a polynomial!

$$\frac{6x}{1-x^3} = \sum_{n=0}^{\infty} 6x(x^3)^n = \sum_{n=0}^{\infty} 6x^{3n+1} = 6x + 6x^4 + 6x^7 + 6x^{10} + \dots$$

$$\int \frac{6x}{1-x^3} dx = \frac{6x^2}{2} + \frac{6x^5}{5} + \frac{6x^8}{8} + \frac{6x^{11}}{11} + \dots + \frac{6x^{3n+2}}{3n+2} + C$$

Ex. 4: If $h'(x) = 1 - 3x + 9x^2 - 27x^3 + \dots$

a) Write the function represented by $h'(x)$ and then write the Maclaurin series for $h(x)$.

h' has repeated multiplication of $-3x$..

$$h' = \sum_{n=0}^{\infty} (-3x)^n \text{ This is geometric so } h' = \frac{1}{1-3x}$$

$$h' = \frac{1}{1-3x}$$

$$h(x) = \int h'(x) dx = \int \sum_{n=0}^{\infty} (-1)^n 3^n x^n dx = \sum_{n=0}^{\infty} \frac{(-1)^n 3^n x^{n+1}}{n+1} + C$$

b) Find the radius of convergence of $h(x)$.

$h(x)$ isn't geometric (h' is)

$$\lim_{n \rightarrow \infty} \left| \frac{3^{n+1} x^{n+2}}{n+2} \cdot \frac{n+1}{3^n x^{n+1}} \right| < 1 \rightarrow \lim_{n \rightarrow \infty} \left| \frac{n+1}{n+2} \cdot 3x \right| < 1$$

as $n \rightarrow \infty$, this goes to 1

$$|3x| < 1 \rightarrow -1 < 3x < 1, \quad -\frac{1}{3} < x < \frac{1}{3} \text{ so } R = \frac{1}{3}$$

← $C=0$ →
-1/3 1/3

Ex. 5:

The Maclaurin series for the function f is given by $f(x) = \sum_{n=2}^{\infty} \frac{(-1)^n (2x)^n}{n-1}$ on its interval of convergence.

(a) Find the interval of convergence for the Maclaurin series of f . Justify your answer.

(b) Show that $y = f(x)$ is a solution to the differential equation $xy' - y = \frac{4x^2}{1+2x}$ for $|x| < R$, where R is the radius of convergence from part (a).

$$a) \lim_{n \rightarrow \infty} \left| \frac{(2x)^{n+1}}{n} \cdot \frac{n-1}{(2x)^n} \right| < 1 \rightarrow \lim_{n \rightarrow \infty} \left| \frac{n-1}{n} \cdot 2x \right| < 1$$

as $n \rightarrow \infty$, this goes to 1

$$|2x| < 1, -1 < 2x < 1, -1/2 < x < 1/2 \quad \text{[not geometric, so test endpoints]}$$

$$x = -1/2: \sum_{n=2}^{\infty} \frac{(-1)^n (-1)^n}{n-1} \quad \boxed{\text{diverges}} \text{ by harmonic series}$$

$$x = 1/2: \sum_{n=2}^{\infty} \frac{(-1)^n (1)^n}{n-1} \quad \boxed{\text{converges}} \text{ by alternating harmonic}$$

$$\boxed{-1/2 < x \leq 1/2}$$

$$b) y' = \sum_{n=2}^{\infty} \frac{(-1)^n n (2x)^{n-1} \cdot 2}{n-1}$$

multiply by x

$$xy' = \sum_{n=2}^{\infty} \frac{(-1)^n n (2x)^{n-1} \cdot 2x}{n-1} = \sum_{n=2}^{\infty} \frac{(-1)^n n (2x)^n}{n-1}$$

$$- y = \sum_{n=2}^{\infty} \frac{(-1)^n (2x)^n}{n-1}$$

$$xy' - y = \sum_{n=2}^{\infty} \frac{(-1)^n n (2x)^n}{n-1} - \sum_{n=2}^{\infty} \frac{(-1)^n (2x)^n}{n-1} = \sum_{n=2}^{\infty} \frac{(-1)^n (n-1) (2x)^n}{n-1}$$

$$xy' - y = \sum_{n=2}^{\infty} (-1)^n (2x)^n = \sum_{n=2}^{\infty} (-2x)^n \xrightarrow{\text{geometric!}} \frac{(-2x)^2}{1 - (-2x)} = \boxed{\frac{4x^2}{1+2x}}$$

AP Calculus II

Notes 9.7

Error Approximation

One of the bigger uses of Taylor polynomials is to approximate values of a transcendental/trigonometric function. This leads us to the idea of the accuracy of the approximation.

$$f(c) = P_n(c) + R_n(c)$$

actual/ exact = polynomial approximation + Remainder

So, $R_n(c) = f(c) - P_n(c)$. This means that the **error** of the approximation can be expressed as:

$$\text{Error} = |f(c) - P_n(c)|$$

error is considered positive even if above or below

We can find this error using one of two methods, which depend on whether the polynomial alternates. We will start off with if the polynomial does alternate:

Alternating Series Remainder:

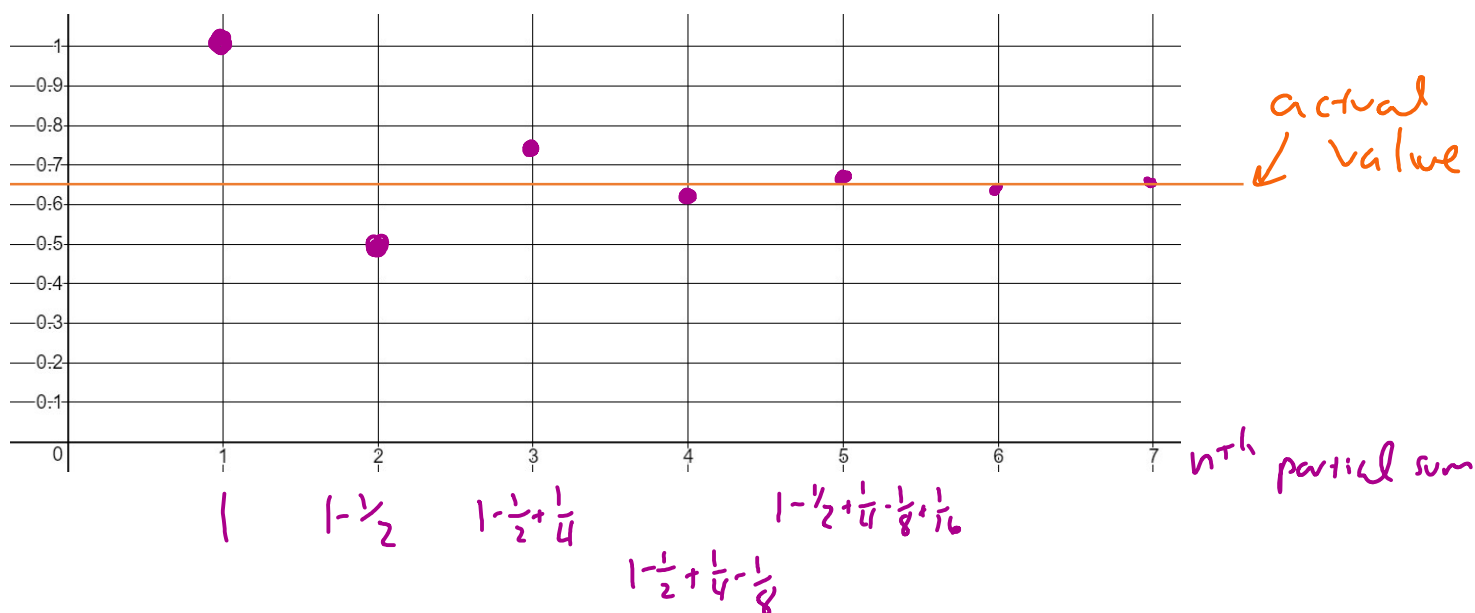
If a series $\sum_{n=1}^{\infty} a_n$ is **alternating, decreasing in absolute value** and $\lim_{n \rightarrow \infty} a_n = 0$, then the series converges.

If you look at the partial sums of the series, it will bounce back and forth around the actual sum. To model this, let's look at the following geometric series:

$$\sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n = 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{16} - \frac{1}{32} + \dots$$

Knowing that this series is geometric, we know that this series adds up to $\frac{1}{1 - (-1/2)} = \frac{2}{2+1} = \boxed{\frac{2}{3}}$ actual value

Let's imagine we didn't know what this series equals, so let's approximate the sum using the first few partial sums.



By introducing the next term in the series, you go from 1 estimate to the next estimate. You also pass through the actual value.

So, if one partial sum is larger than the true value, then subtracting the next term will cause the partial sum to be smaller and thus, less than the true value. Then, adding the next term will cause the partial sum to be larger and so on... The error is the difference between any of the partial sums and the actual value, but by adding (or subtracting) the next term, the partial sum will go past the actual value. Therefore, the **maximum error** is less than or equal to the **first (next) neglected term** in the series.

Again, this applies to **convergent alternating series that decrease in absolute value to 0.**

↙ looks geometric!

Ex. 1: Given $g(x) = \frac{6}{2+x}$, write the first four nonzero terms for the Maclaurin polynomial and the general term for $g(x)$. Using this polynomial, approximate $g(1)$ and determine the maximum error.

$$g = \frac{6}{2+x} = \frac{3}{1+x/2} = \frac{3}{1-(-x/2)} = \sum_{n=0}^{\infty} 3(-x/2)^n = 3 - \frac{3x}{2} + \frac{3x^2}{4} - \frac{3x^3}{8} + \dots$$

$$P_4(x) = 3 - \frac{3}{2}x + \frac{3}{4}x^2 - \frac{3}{8}x^3 + \frac{3}{16}x^4 \quad g(1) \approx P_4(1) = 1.9125$$

Since g is a convergent alternating series that decreases in absolute value to 0...

$$|g(1) - P_4(1)| \leq |\text{next term}| \rightarrow |g(1) - 1.9125| \leq \left| \frac{-3x^5}{32} \right|_{x=1} \Rightarrow 0.09375$$

$g(1) = \frac{6}{3} = 2$ Actual error = 0.0875

Ex. 2: Given $f(x) = \sin x$, write the fifth degree Maclaurin polynomial and the general term for $f(x)$. Using this polynomial, approximate $\sin(0.1)$ and determine the maximum error.

$$\sin x \approx x - \frac{x^3}{3!} + \frac{x^5}{5!} = P_5(x), \quad \text{General term: } \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\sin(0.1) \approx P_5(0.1) = 0.09983341666667$$

Since f is a convergent, alternating series that decreases in absolute value to 0,

$$|\sin 0.1 - P_5(0.1)| \leq \left| \frac{-x^7}{7!} \right|_{x=0.1} \quad \text{max error} = 1.9841 \times 10^{-11}$$

actual approximate

$\sin 0.1 = 0.09983341664$ ← 11th digit

Ex. 3: Given $f(x) = e^{-2x}$,

a) Write the third degree Maclaurin polynomial $P_3(x)$ for $f(x)$.

either build f or develop from general form of poly.

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$e^{-2x} \approx P_3(x) = 1 - 2x + \frac{4x^2}{2!} - \frac{8x^3}{3!}$$

replace x with $-2x$

b) Using $P_3(x)$, approximate $f(0.5)$. Then, show that $f(0.5)$ differs from $P_3(0.5)$ by less than $\frac{1}{20}$.

$$f(0.5) = e^{-1} \approx P_3(0.5) = 1 - 1 + \frac{1}{2} - \frac{1}{6} = 0.33333333$$

since f is a convergent, alternating series that decreases in absolute value to 0

$$|e^{-1} - 0.333| \leq \left| \frac{16(0.5)^4}{4!} \right| = \frac{16(\frac{1}{2})^4}{4!} = \frac{16(\frac{1}{16})}{4!} = \frac{1}{24} < \frac{1}{20} \checkmark$$

Ex. 4: The Taylor series for f about $x = 2$ converges to f for $-1 \leq x \leq 5$. The n th-degree Taylor

polynomial for f about $x = 2$ is given by $P_n(x) = \sum_{k=1}^n (-1)^k \frac{(x-2)^k}{3^k(k^2+1)}$. What is the smallest

degree n of the Taylor polynomial that guarantees that $|f(3) - P_n(3)| \leq \frac{1}{100,000}$?

Since f is a convergent, alternating series that decreases in absolute value to 0, the max error is $\frac{1}{100,000}$, which is the value of the next term.

if $n=4 \rightarrow 5^{\text{th}}$ degree term is max error $\rightarrow \frac{(3-2)^5}{3^5(25+1)} = \frac{1}{6318} \dots \text{not} \leq \frac{1}{100,000}$

if $n=6 \rightarrow 7^{\text{th}}$ degree term is max error $\rightarrow \frac{(3-2)^7}{3^7(49+1)} = \frac{1}{109350} \leq \frac{1}{100,000}$

if $n=5 \rightarrow 6^{\text{th}}$ degree term is max error $\rightarrow \frac{(3-2)^6}{3^6(36+1)} = \frac{1}{26,973}$

So $n=6$

Ex. 5: The Taylor series about $x = 4$ for a certain function f converges to $f(x)$ for all x in the interval of convergence. The n th derivative of f at $x = 4$ is given by

$$f^{(n)}(4) = \frac{(-1)^n(n+1)!}{2^n} \quad \text{and } f(4) = 1$$

a) Write the third-degree Taylor polynomial for $f(x)$ centered at $x = 4$ and find the general term.

can't build from known series ...

$$P_3(x) = f(4) + \frac{f'(4)}{1!}(x-4) + \frac{f''(4)}{2!}(x-4)^2 + \frac{f'''(4)}{3!}(x-4)^3$$

$$f(4) = 1, \quad f'(4) = -\frac{2!}{2}, \quad f''(4) = \frac{3!}{4}, \quad f'''(4) = -\frac{4!}{8}$$

$$P_3(x) = 1 - \frac{2!}{1!}(x-4) + \frac{3!}{2!}(x-4)^2 - \frac{4!}{3!}(x-4)^3$$

$$P_3(x) = 1 - \frac{2}{1}(x-4) + \frac{3}{2}(x-4)^2 - \frac{4}{1}(x-4)^3$$

general term = $\frac{(-1)^n(n+1)!}{2^n n!} (x-4)^n = \frac{(-1)^n(n+1)}{2^n} (x-4)^n$

b) Find the coefficient of the 10th degree term of the Taylor series for f centered at $c = 4$.

if general term is $\frac{(-1)^n(n+1)}{2^n} (x-4)^n$

$n=10 \rightarrow \frac{11}{2^{10}} (x-4)^{10}$ so $\frac{11}{2^{10}}$

c) Show that the third-degree Taylor polynomial approximates $f(5)$ with an error less than $\frac{1}{3}$.

Since f is a convergent, alternating series that decreases in absolute value to 0,

$$|f(5) - P_3(5)| \leq \left| \begin{matrix} 4^{\text{th}} \text{ degree} \\ \text{term at } x=5 \end{matrix} \right| \rightarrow \frac{5}{16} (5-4)^3 = \frac{5}{16} < \frac{1}{3}$$

Yes ✓

What if the series is not alternating?

If the series is not alternating, you must use the Lagrange Error in which the maximum error will occur at some value z , such that z is in between x and c .

center $\rightarrow c \leq z \leq x \leftarrow$ value used to find approximate

Lagrange Error Bound/Taylor's Theorem:

For $f(x) = f(c) + \frac{f'(c)}{1!}(x-c) + \frac{f''(c)}{2!}(x-c)^2 + \frac{f'''(c)}{3!}(x-c)^3 + \dots + \frac{f^{(n)}(c)}{n!}(x-c)^n + R_n(x)$

where the maximum value of the remainder is $R_n(x) = \left| \frac{f^{(n+1)}(z)}{(n+1)!}(x-c)^{n+1} \right|$.
 $\leftarrow$ still uses next term

This is very similar to the Alternating Series Remainder in that they both use the next term in the series. However when using Lagrange Error, you must find or use reasoning to determine the z value that will make the next derivative the maximum value possible. This will make the remainder the largest possible and will truly represent the maximum error.

Ex. 6: Given the function $f(x) = 3e^x$,

- a) Write the fourth-degree Maclaurin polynomial for $f(x)$ and the general term. Use this fourth-degree polynomial to approximate $3e^2$.

$$f(x) \approx P_4(x) = 3 + 3x + \frac{3x^2}{2!} + \frac{3x^3}{3!} + \frac{3x^4}{4!} \quad \text{general term: } \frac{3x^n}{n!}$$

$$f(z) = 3e^2 \approx P_4(z) = 21$$

- b) Find the maximum error of this approximation to the actual value of $3e^2$.

f is not alternating, so must use Lagrange

$$\left| \underset{\substack{\uparrow \\ \text{actual}}}{f(z)} - \underset{\substack{\uparrow \\ \text{approximate}}}{21} \right| \leq \left| \frac{f^{(5)}(z)}{5!} x^5 \right|_{x=2} \quad \text{where } z \text{ is a number between } 0 \leq z \leq 2 \text{ that will make } f^{(5)}(x) \text{ a maximum}$$

$f^{(5)}(x) = 3e^x$, which is always increasing. So therefore, $z=2$ would make $f^{(5)}(x)$ a max.

$$\text{max error} = \frac{f^{(5)}(2)}{5!}(2)^5 = \frac{3e^2}{5!}(2)^5 = 5.9112$$

Ex. 7: The function $f(x) = \frac{1}{(1-x)^2}$ can be expressed as the series $\sum_{n=1}^{\infty} nx^{n-1}$ for all values of x within its interval of convergence.

a) Find the interval of convergence of $f(x)$.

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)x^n}{n x^{n-1}} \right| < 1 \rightarrow \lim_{n \rightarrow \infty} \left| \frac{n+1}{n} \cdot x \right| < 1$$

as $n \rightarrow \infty$, this goes to 1

$|x| < 1$
 $-1 < x < 1$

check endpoints!
 $x = -1: \sum n(-1)^{n-1}$ diverges by n^{th} term test for divergence
 $x = 1: \sum n(1)^{n-1}$ diverges by n^{th} term test for divergence

$(-1, 1)$

b) Use the first 4 nonzero terms of f to approximate the value of $f\left(\frac{1}{3}\right)$. Show that this approximation differs from $f\left(\frac{1}{3}\right)$ by less than $\frac{3}{4}$.

$$f \approx 1 + 2x + 3x^2 + 4x^3 = P_3(x), \quad P_3\left(\frac{1}{3}\right) = 2.148148$$

Not alternating. So must maximize the $f^{(4)}$ & next derivative

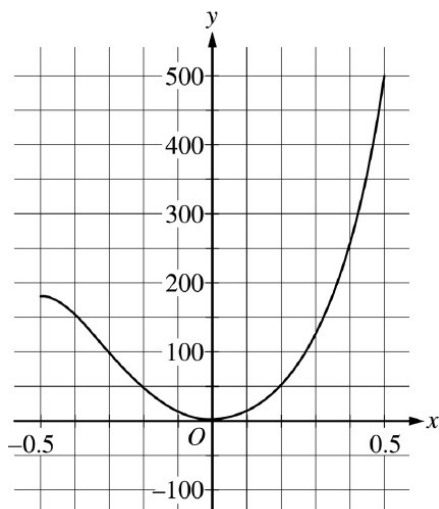
$$\begin{aligned} f &= (1-x)^{-2} \\ f' &= +2(1-x)^{-3} \\ f'' &= 6(1-x)^{-4} \\ f''' &= 24(1-x)^{-5} \\ f^{(4)} &= 120(1-x)^{-6} \end{aligned}$$

So $f^{(4)}(x) = \frac{120}{(1-x)^6}$. So what value of z , where $0 \leq z \leq \frac{1}{3}$, will make $f^{(4)}$ the greatest value. As x increases, the denominator gets smaller. This cause the $f^{(4)}$ to get larger ... so $z = \frac{1}{3}$ is the max!

$$\left| f\left(\frac{1}{3}\right) - P_3\left(\frac{1}{3}\right) \right| \leq \left| \frac{f^{(4)}\left(\frac{1}{3}\right)}{4!} \left(\frac{1}{3}\right)^4 \right| \rightarrow \frac{\frac{120}{(1-\frac{1}{3})^6}}{4!} \left(\frac{1}{3}\right)^4 = 0.703 < \frac{3}{4}$$

✓

Ex. 8: Let the function f be defined as $f(x) = x^2 e^{x^2}$. The graph of $f^{(7)}$, the seventh derivative of f , can be seen below.



- a) Write the first four nonzero terms and general term for the Maclaurin series for e^x . Then, use this series to write the first four nonzero terms and general term for the Maclaurin series for f .

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$$

$$e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \dots + \frac{x^{2n}}{n!}$$

$$f = x^2 e^{x^2} = x^2 + x^4 + \frac{x^6}{2!} + \frac{x^8}{4!} + \dots + \frac{x^{2n+2}}{n!}$$

replace x with x^2

multiply by x^2

- b) Let $P_6(x)$ be the sixth-degree Taylor polynomial for f about $x = 0$. Find $P_6\left(\frac{1}{2}\right)$, then use the Lagrange error bound along with the graph to find an upper bound on $\left|P_6\left(\frac{1}{2}\right) - f\left(\frac{1}{2}\right)\right|$.

$$P_6 = x^2 + x^4 + \frac{x^6}{2!}$$

$$\rightarrow P_6\left(\frac{1}{2}\right) = 0.3203125$$

Since f is not alternating, use Lagrange!

$$\left|f\left(\frac{1}{2}\right) - P_6\left(\frac{1}{2}\right)\right| \leq \left|\frac{f^{(7)}(z)}{7!} x^7\right|$$

$$0 \leq z \leq \frac{1}{2}$$

• According to graph of $f^{(7)}$, the max of $f^{(7)}$ from $[0, \frac{1}{2}]$ is 500

$$\text{So, Max error} = \frac{500}{7!} \left(\frac{1}{2}\right)^7 = 0.000775$$

2011 AP[®] CALCULUS BC FREE-RESPONSE QUESTIONS (Form B)

6. Let $f(x) = \ln(1 + x^3)$.

- (a) The Maclaurin series for $\ln(1 + x)$ is $x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + (-1)^{n+1} \cdot \frac{x^n}{n} + \cdots$. Use the series to write the first four nonzero terms and the general term of the Maclaurin series for f .
- (b) The radius of convergence of the Maclaurin series for f is 1. Determine the interval of convergence. Show the work that leads to your answer.
- (c) Write the first four nonzero terms of the Maclaurin series for $f'(t^2)$. If $g(x) = \int_0^x f'(t^2) dt$, use the first two nonzero terms of the Maclaurin series for g to approximate $g(1)$.
- (d) The Maclaurin series for g , evaluated at $x = 1$, is a convergent alternating series with individual terms that decrease in absolute value to 0. Show that your approximation in part (c) must differ from $g(1)$ by less than $\frac{1}{5}$.

WRITE ALL WORK IN THE EXAM BOOKLET.

END OF EXAM

2005 AP[®] CALCULUS BC FREE-RESPONSE QUESTIONS (Form B)

3. The Taylor series about $x = 0$ for a certain function f converges to $f(x)$ for all x in the interval of convergence. The n th derivative of f at $x = 0$ is given by

$$f^{(n)}(0) = \frac{(-1)^{n+1}(n+1)!}{5^n(n-1)^2} \text{ for } n \geq 2.$$

The graph of f has a horizontal tangent line at $x = 0$, and $f(0) = 6$.

- (a) Determine whether f has a relative maximum, a relative minimum, or neither at $x = 0$. Justify your answer.
 - (b) Write the third-degree Taylor polynomial for f about $x = 0$.
 - (c) Find the radius of convergence of the Taylor series for f about $x = 0$. Show the work that leads to your answer.
-

WRITE ALL WORK IN THE TEST BOOKLET.

END OF PART A OF SECTION II

2014 AP[®] CALCULUS BC FREE-RESPONSE QUESTIONS

6. The Taylor series for a function f about $x = 1$ is given by $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{2^n}{n} (x-1)^n$ and converges to $f(x)$ for $|x-1| < R$, where R is the radius of convergence of the Taylor series.
- (a) Find the value of R .
- (b) Find the first three nonzero terms and the general term of the Taylor series for f' , the derivative of f , about $x = 1$.
- (c) The Taylor series for f' about $x = 1$, found in part (b), is a geometric series. Find the function f' to which the series converges for $|x-1| < R$. Use this function to determine f for $|x-1| < R$.
-

STOP

END OF EXAM

2009 AP[®] CALCULUS BC FREE-RESPONSE QUESTIONS (Form B)

6. The function f is defined by the power series

$$f(x) = 1 + (x + 1) + (x + 1)^2 + \cdots + (x + 1)^n + \cdots = \sum_{n=0}^{\infty} (x + 1)^n$$

for all real numbers x for which the series converges.

- (a) Find the interval of convergence of the power series for f . Justify your answer.
- (b) The power series above is the Taylor series for f about $x = -1$. Find the sum of the series for f .
- (c) Let g be the function defined by $g(x) = \int_{-1}^x f(t) dt$. Find the value of $g\left(-\frac{1}{2}\right)$, if it exists, or explain why $g\left(-\frac{1}{2}\right)$ cannot be determined.
- (d) Let h be the function defined by $h(x) = f(x^2 - 1)$. Find the first three nonzero terms and the general term of the Taylor series for h about $x = 0$, and find the value of $h\left(\frac{1}{2}\right)$.

WRITE ALL WORK IN THE EXAM BOOKLET.

END OF EXAM

2007 AP[®] CALCULUS BC FREE-RESPONSE QUESTIONS (Form B)

5. Consider the differential equation $\frac{dy}{dx} = 3x + 2y + 1$.

- (a) Find $\frac{d^2y}{dx^2}$ in terms of x and y .
- (b) Find the values of the constants m , b , and r for which $y = mx + b + e^{rx}$ is a solution to the differential equation.
- (c) Let $y = f(x)$ be a particular solution to the differential equation with the initial condition $f(0) = -2$. Use Euler's method, starting at $x = 0$ with a step size of $\frac{1}{2}$, to approximate $f(1)$. Show the work that leads to your answer.
- (d) Let $y = g(x)$ be another solution to the differential equation with the initial condition $g(0) = k$, where k is a constant. Euler's method, starting at $x = 0$ with a step size of 1, gives the approximation $g(1) \approx 0$. Find the value of k .
-

6. Let f be the function given by $f(x) = 6e^{-x/3}$ for all x .

- (a) Find the first four nonzero terms and the general term for the Taylor series for f about $x = 0$.
- (b) Let g be the function given by $g(x) = \int_0^x f(t) dt$. Find the first four nonzero terms and the general term for the Taylor series for g about $x = 0$.
- (c) The function h satisfies $h(x) = kf'(ax)$ for all x , where a and k are constants. The Taylor series for h about $x = 0$ is given by

$$h(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots.$$

Find the values of a and k .

WRITE ALL WORK IN THE EXAM BOOKLET.

END OF EXAM

2013 AP[®] CALCULUS BC FREE-RESPONSE QUESTIONS

6. A function f has derivatives of all orders at $x = 0$. Let $P_n(x)$ denote the n th-degree Taylor polynomial for f about $x = 0$.
- (a) It is known that $f(0) = -4$ and that $P_1\left(\frac{1}{2}\right) = -3$. Show that $f'(0) = 2$.
- (b) It is known that $f''(0) = -\frac{2}{3}$ and $f'''(0) = \frac{1}{3}$. Find $P_3(x)$.
- (c) The function h has first derivative given by $h'(x) = f(2x)$. It is known that $h(0) = 7$. Find the third-degree Taylor polynomial for h about $x = 0$.
-

STOP

END OF EXAM